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Satellite remote sensing to improve source water quality monitoring: A water utility's perspective

John Lioumbas^{*}, Aikaterini Christodoulou, Matina Katsiapi, Nikoletta Xanthopoulou, Panagiota Stournara, Thomas Spahos, Georgia Seretoudi, Alexandros Mentas, Nopi Theodoridou

Thessaloniki Water Supply & Sewerage Co S.A., Thessaloniki, Greece

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ABSTRACT

Polyphytos Reservoir provides the city of Thessaloniki with efficient and safe drinking water after appropriate treatment at the Thessaloniki Drinking Water Treatment Plant (TDWTP). This study evaluates the feasibility of utilizing open access satellite imagery and analysis tools to continuously, rapidly, at a low cost and precisely evaluate the water quality of large reservoirs like Polyphytos Reservoir. Thereby, more than 300 satellite images were analyzed, spanning a 9-year period from 2013 to 2022, using specific algorithms designed for determining surface water quality, such as Se2WaQ, NDWI and Oil Spill Index (OSI). These algorithms were used to estimate chlorophyll concentration (Chl-a), turbidity values (Tur) and hydrocarbon (OSI) presence. The generated results are compared with data records regarding the reservoir's water level and chlorophyll and turbidity measurements in the reservoir and at the inlet of the TDWTP. This study shows that the examined satellite survey tools can effectively identify unanticipated formations on the surface of the reservoir and provide a quantitative assessment of the surface water quality using in-situ sampling for algorithms validation. The findings of the study were used to create a workflow that can serve as a guide for end-users interested in a low-cost, quick, continuous, and accurate surface water quality assessment of large reservoirs, using satellite remote sensing tools. It is anticipated that the information provided in this study would be useful to end-users community.

1. Introduction

The city and the suburbs of Thessaloniki receive part (currently 50% but this amount is estimated to increase up to 100% in near future) of its drinking water from the river Aliakmonas through an extended water route. This route includes Aliakmon River (322 km), a complicated system of artificial reservoirs (the largest is Polyphytos covering an area of more than 75 km²), an open, free flow, water transfer channel (more than 50 km long) and a sealed pipe (near 10 km long). The water is received by the Thessaloniki Drinking Water Treatment Plant (TDWTP) and treated under the responsibility of Thessaloniki Water Supply & Sewerage Co S.A., (EYATH SA) to fulfill the drinking water quality standards according to the Drinking Water Directive (EU, 2020/2184) and the respective National legislation. However, in recent years, there have been various unexpected pollution incidents along the above water path, such as increased turbidity levels, traces of chlorophyll-a, presence of cyanobacteria, and hydrocarbons, that have negatively impacted the quality of the water flowing into the TDWTP, requiring unforeseen adjustments to the planned treatment processes. The use of very high frequency water quality monitoring along the above-mentioned water pathway would provide the TDWTP operators

^{*} Corresponding author.

E-mail address: ilioumbas@eyath.gr (J. Lioumbas).

the necessary time to take effective preventive measures in case of pollution emergencies (Papenfus et al. 2020). While this is crucial also due to the inherent risk posed by the variety and quantity of anthropogenic pressures (e.g. the Polyphytos Reservoir is used for hydroelectric power, fishing, sports, irrigation etc), it can be highly challenging and resource-intensive to achieve using traditional monitoring methods (Birk and Ecke 2014).

The last four decades, Satellite Remote Sensing Techniques (SRST) are playing an increasing role in assessing water quantity and quality of inland waters, largely due to technological advances, including instrument/sensor and algorithm/image processing improvements (Reif, 2011). Remote sensing data and products may provide synoptic, spatio-temporal views of changes in water characteristics and their integration in monitoring programs can contribute to a better understanding of water quality, although they should not completely substitute *in situ* measurements (Caballero et al., 2022) which are more accurate, precise, and specific. Regarding the water quality monitoring of lakes, SRST have been widely used to measure various lake properties such as water transparency (e.g. turbidity, colored dissolved organic matter, turbidity), biota (chlorophyll-a, cyanobacteria, submerged and emerged aquatic vegetation), bathymetry and water temperature (surface temperature) (Dornhofer and Oppelt 2016). Nevertheless, stakeholders and end-user community have been slow to embrace satellite-derived measurements to date (Kallio, 2007) since many management and policy professionals lack a base knowledge in remote sensing since it has not yet been incorporated into professional training curriculums (Greb et al. 2018). Moreover, it can be difficult for an end-user to integrate satellite image analysis tools, which may involve the use of complex algorithms and specialized scientific expertise into their daily operational practices. Specifically, the use of satellite remote sensing tools (SRST) for water quality monitoring could be a challenging task due to a continuous need for calibration that takes input from traditional sampling methods and field surveying (Hajigholizadeh et al., 2016), the difficulty of atmospheric correction for inland waters (Ogashawara et al., 2021), and the variability of relationships between reflectance and water quality across different regions (Barrett and Frazier, 2016). These challenges can be addressed through the development of special algorithms and machine learning approaches, but the high sophistication of these tools can make them impractical for standard operating procedures of a water utility (Barrett and Frazier, 2016).

In recent years, EYATH SA has implemented an internal company initiative aimed at evaluating the feasibility of using satellite remote sensing techniques (SRSTs) to monitor the surface water quality of the Polyphytos Reservoir and investigate any potential correlations between the reservoir's water quality and that of the water entering the Thessaloniki Drinking Water Treatment Plant (TDWTP). This was achieved by leveraging open-access, online cloud platforms offering high spatial resolution satellite imagery, such as the Sentinel Hub EO Browser (<https://apps.sentinel-hub.com>), combined with freely available custom scripts. The objective was to determine the capacity to identify specific anomalies on the surface of the Polyphytos Reservoir (Lioumbas et al., 2023a) and monitor fluctuations in the reservoir's level due to dam management (Lioumbas et al., 2023b). The preliminary findings suggest that this cost-effective approach offers significant benefits for monitoring the water quality of large Reservoirs, making it accessible to end-users with limited expertise in the field, such as water utilities.

Specifically, the Sentinel Hub EO Browser platform allows the access of high-resolution data from multiple satellites sources (e.g. Sentinel-1-2, Landsat 8-9 etc.) with almost no loading time. Moreover, the Sentinel Hub EO Browser includes algorithms for the atmospheric correction (Sentinel-2 Correction; Sen2Cor, developed by ESA specifically for Sentinel-2 Level 2A land product (Müller-Wilm et al., 2013) that are suitable for inland waters (Warren et al. 2019). Furthermore, the satellite images could be further processed by employing free access custom algorithms that seem to be a simpler, yet efficient, way to estimate the water quality and are especially dedicated to the evaluation of various parameters. More specifically, the algorithm Sentinel-2 Water Quality Script (Se2WaQ) which has been developed and evaluated for inland water quality monitoring (Potes et al. 2018; Viso-Vázquez et al. 2021) are utilized in this study for the Chl-a and turbidity monitoring. The algorithm was originally developed for water quality monitoring in the Alqueva Reservoir, Portugal, which shares similarities with the Polyphytos Reservoir¹. The Oil Spill Index is employed as suggested by Rajendran et al. (2021) for oil spill detection using Sentinel-2 data. NDWI was first proposed by McFeeters (1996) to detect surface waters in wetland environments and to allow for the measurement of surface water extent. These indices were selected because they are referenced in literature as suitable for monitoring water quality in inland water, due to their simplicity and availability of pre-existing scripts, which can be utilized by non-expert end-users.

Concerning the use of SRST tools to evaluate the water quality of lakes in Greece, there are only few studies available (e.g. Perivolioti et al. 2016; Kontopoulou et al., 2017; Bresciani et al. 2020). Markogianni et al. (2020) employed Chl-a as an index to assess the water quality of 50 different inland water bodies in Greece. Markogianni et al. (2020) compared EO data with in-situ measurements of Total Phosphorus and Secchi depth to model the Greek lakes water quality. The above studies suggest that a satellite-based methodology could be a promising tool for "public authorities to guide and eventually manage sustainably public safety decisions and effective protection measures" (Markogianni et al. 2020). However, for Polyphytos Reservoir there are no studies that systematically evaluate its environmental state by SRSTs.

Numerous studies have employed SRSTs to evaluate distinct water quality parameters in inland waters, including Harmful Algal Blooms (HAB; Wynne et al., 2010), elevated turbidity (Matthews, 2011), and the presence of hydrocarbons (García-Pineda et al., 2013). However, only a few studies (e.g., Laneve et al. 2021) combine different SRST algorithms to discriminate the origins of observed phenomena. This is not unexpected, given the complexity of interrelated phenomena, the inherent limitations of SRSTs, and the challenge of differentiating between a range of potential environmental stress parameters. However, the needs of an end-user who

¹ Both being artificial reservoirs created by dam construction, both being situated in Mediterranean climates characterized by hot, dry summers and mild, wet winters, and both potentially experiencing eutrophication issues due to agricultural activities in their respective catchment areas, leading to elevated nutrient levels and potential algal blooms. The script can be found at: Se2WaQ - Sentinel-2 Water Quality Script | Sentinel Hub custom scripts ([sentinel-hub.com](https://apps.sentinel-hub.com)) (author of the script: Nuno Sidónio Andrade Pereira).

aims to develop strategies and make quick (in emergency situations) or long-term decisions on managing water bodies, are often focused on gaining a better understanding of the factors that affect surface water quality (both spatially and temporally) and identifying the source of formations in a large water body, rather than solely determining the precise value of a single parameter. In particular for Polyphytos Reservoir, the degradation of water quality in the inlet of the TDWTP may be attributed to various anthropogenic pressures in the vicinity of the reservoir, which could negatively affect its integrity. Therefore, this study aims to evaluate the effectiveness of Satellite Remote Sensing Techniques (SRSTs) in offering insights into the possible sources of observed anomalies on the Reservoir surface. This assessment is conducted alongside an analysis of environmental factors that could be linked to elevated chlorophyll concentrations or increased turbidity levels, such as seasonal temperature variations, reservoir bathymetry, and water level fluctuations due to dam management.

This study aims to showcase how end-users, including water utilities, can take advantage of affordable and user-friendly satellite monitoring tools. These tools can assist in pinpointing factors that lead to the decline in water quality and the potential appearance of unexpected surface formations in inland water bodies. Moreover, the research intends to explore the extent to which these factors influence the reservoir's environmental status over time. The specific objectives of the study are as follows:

1. Examine the integration of these tools into water utility processes, enhancing monitoring and early warning services, particularly in locations with sparse in-situ measurements.
2. Develop a step-by-step process for end users possessing limited or moderate expertise in satellite remote sensing data analysis, enabling them to effectively evaluate the water quality of large reservoirs.

To achieve the stated objectives, this study employs the following steps: First, a detailed description of the "Materials and Methods" is provided, including information on the study area, existing water quality monitoring strategies in Polyphytos Reservoir, regional meteorology and climate, the methods used to calculate water quality indices and measure the reservoir's water level, and the approach used to analyze and evaluate the results. The study is completed with sections on Results and Conclusions.

2. Materials and Methods

2.1. Study area

Polyphytos Reservoir was built in 1975, following the construction of a 112-m-high dam in the vicinity of Polyphytos Village in Kozani Province, West Macedonia, Greece. The water level in the reservoir is affected mainly by the operation of the hydropower plant and the irrigation requirements of the surrounding agricultural region. The Ilarion and Polyphytos dams are in the southern and northern regions of the reservoir, respectively. As shown in Fig. 1, the reservoir's depth is shallow in the southern area and significantly increases in the northern region. Moreover, Fig. 1 shows that Polyphytos Reservoir is surrounded by settlements and small sub-basins, which are formed due to the intense uplift of the area. This specific morphology also favors the transport of pollutants within the reservoir (Baklavaridis, 2018). Potential sources of pollution in the reservoir (indicated with magenta dots in Fig. 1) may include livestock and agricultural activities, small milk and meat processing industries, the operation of a thermal power plant, and coal mining in the areas between the cities of Kozani and Ptolemaida, as well as two coal mines on the eastern side of the Reservoir. In addition, many cities and villages in the area discharge their untreated municipal wastewater into the main reach of the Aliakmon River upstream of the reservoir, as well as into various torrents that flow directly into the reservoir (Gikas et al. 2008; Zotou et al., 2019; Baklavaridis, 2018). Currently, Polyphytos Reservoir is part of the National Water Monitoring Network (in accordance with the WFD, 2000/60/EU implementation in Greece) and is also being monitored by the Public Power Corporation (PPC SA) and EYATH SA.

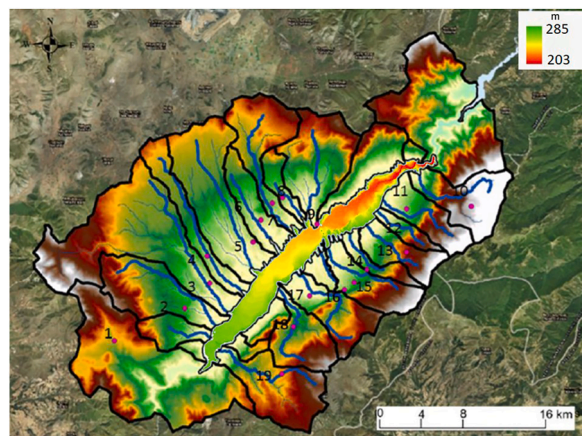


Fig. 1. Polyphytos reservoir, showing the hydrographic network (black lines), the main watercourses (blue lines), the points corresponding to the main settlements around it (Baklavaridis, 2018) along with the bathymetric data of the Polyphytos lake (data obtained from: Technical report, 2011). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.2. Meteorological and water quality data

The data and information used in this study on water quality, including turbidity at the entrance of the TDWTP and incidents such as phytoplankton blooms and oil spills, were obtained from the monitoring program database of EYATH SA. This database includes information from Polyphytos Reservoir and various sampling points along the Aliakmon River up to the point where the water enters the TDWTP, including both *in situ* sampling and online measurements. Aiming to further validate the capability of the Chl-a and Tur algorithms in describing the water quality of Polyphytos Reservoir and better understand Polyphytos Reservoir environmental conditions, EYATH SA initiated in-situ sampling campaigns at dates that are close to the passages of Sentinel-2, over Polyphytos Reservoir. Water samples (2 L per sampling point) were collected at 30 sampling points along the reservoir for the determination of Chl-a, turbidity and other physicochemical parameters (e.g. water temperature). Chlorophyll *a* was measured spectrophotometrically according to the method of Jeffrey and Humphrey (1975) based on the extraction with 90% acetone. Turbidity was measured in the lab using a turbidimeter (TU5200 HACH). The identification of phytoplankton (and cyanobacterial) species and cyanobacterial biovolume determination were carried out as previously described (Katsiapi et al. 2016).

The various meteorological data have been obtained from the Velvento weather station from the www.meteo.gr depository and shows that during the study period the colder month was January (-7.8°C , 2019), and the warmer month was July (28.5°C , 2020) and the average temperature in all four years was $14.3\text{--}14.4^{\circ}\text{C}$ (Supplementary material A). The annual amount of rainfall in the area is not very high (annual average for years 2018–2021 is 1.76 mm, st. dev.: 0.538 mm). The highest amount of rainfall is recorded from October to May (wet period) but severe rainfall incidents that exceed 30 mm per day may also happen during summer. From January to April, South and North directed winds prevail with high intensity (maximum value: 54.7 Km/h).

2.3. Remote sensing data

The satellite data used in this study were obtained from the Sentinel Hub EO browser, an online platform that provides access to the entire archive of earth observation satellite imagery. The browser features user-friendly visualization and online scripting in JavaScript (Sinergise, 2019). The data used in this study were sourced from the Landsat-8 (L8) and Sentinel-2 (S-2A) satellites. Landsat-8 has a spatial resolution of 30 meters at ground level and acquires images every 16 days. Sentinel-2 (S-2A) has a significant potential for remote sensing of inland water bodies as it combines the high spatial resolution of traditional satellite sensors (e.g., Landsat and SPOT) with the increased temporal and spectral resolutions required for aquatic remote sensing (Laneve et al. 2021). The Level-2A processing includes a Scene Classification and an Atmospheric Correction applied to Top-Of-Atmosphere (TOA) Level-1C orthoimage products. The main output of Level-2A processing which has been used in this study is an atmospherically corrected orthoimage Surface Reflectance product. The Sentinel-2 (S-2A) satellite orbits have a 5-day revisit time in paired operations, with image acquisitions taking place around 9:29 UTC (i.e., 11:29 and 12:29 local time in winter and summer, respectively). Sentinel-2 carries a multi-spectral instrument (MSI) that acquires reflected solar radiance in 13 bands between 440 and 2200 nm spectral wavelength. The spatial resolution is 10 meters for the red, green, blue, and near-infrared bands, 20 meters for six more bands in the NIR (705–865 nm) and short-wave infrared (SWIR, 1600–2200), and 60 meters for three other bands (443, 940, and 1375 nm) (Gascon et al. 2016).

2.3.1. Reservoir water level fluctuations

Satellite altimetry and imagery are now widely used together to calculate, in near real time, the water content of lakes and Reservoirs worldwide (Cretaux et al. 2016; Rodell et al. 2018). In the present study satellite images (LandSat 8–9 L2; false color based on the combination of bands 5, 4 and 3) along with bathymetric data are employed to obtain historical data concerning the evolution of Polyphytos Reservoir water level from 2013 till nowadays. The variations in reservoir water level are calculated by over posing satellite imagery obtained at known dates with bathymetric maps (Technical report, 2011) of Polyphytos Reservoir (a typical example of the images sequence overlap over a nine-year period can be found in Supplementary Material B). The generated progress of water level fluctuations is further validated with the water level fluctuations measured in the dam by the Public Power Corporation (PPC).

2.3.2. Surface water quality assessment and algorithm validation

A comprehensive dataset consisting of over 300 Sentinel-2 and Landsat-8 satellite images, covering a nine-year period, was initially processed using the NDWI algorithm to identify distinct formations (anomalies) on the Reservoir's surface. These features were subsequently classified through an exhaustive visual inspection based on true or/and enhanced color satellite imagery and their physical attributes, such as size, shape, color, and intensity. For further surface water quality analysis, the Se2WaQ – Sentinel-2 Water Quality Script (Chl-a and Turbidity algorithms) and the OSI - Oil Spill Index were employed. Numerous empirical models have been proposed in the literature for modeling chlorophyll-a (Gunawardana et al., 2022; Germán et al., 2021) and turbidity (Maimouni et al., 2022). For this study, the selected chlorophyll-a (Chl-a) and turbidity algorithms, validated specifically for inland water bodies as recommended by Potes et al. (2018), were applied:

$$\text{Chl} - a = 4.23 \left(\frac{560\text{nm}}{442.5\text{nm}} \right)^{3.94} [\text{mg m}^{-3}] \quad (2.1)$$

$$\text{Tur} = 8.93 * \frac{560\text{nm}}{412.5\text{nm}} - 6.39 [\text{NTU}] \quad (2.2)$$

The Oil Spill Index (OSI), developed and validated by Rajendran et al. (2021), was also used to map marine oil spills. The OSI algorithm employs the B3, B4, and B2 bands on Sentinel-2:

$$OSI = \left(\frac{B03 + B04}{B02} \right) \quad (2.3)$$

Upon application of these equations, satellite images were produced. These images were then juxtaposed with same-day in-situ measurements of Chl-a and turbidity, which were timed to coincide with the satellite's passage. This comparison enabled the recalibration of the parameters of Eqs. (2.1) and (2.2) to better align with the in-situ measurements.

An iterative methodology was employed, which involved the systematic refinement of algorithm parameters. This process was continued until the mean absolute difference between the algorithm-derived values and the in-situ measurements was reduced to less than $\pm 2.5\%$ of the initial difference, a criterion defined as the point of 'convergence'. Following this calibration process, the resulting satellite images were further validated using an independent set of in-situ measurements. These validation measurements were collected on a different day to ensure that the calibrated algorithms could produce reliable results under different conditions. Given that the 'convergence' remained within $\pm 2.5\%$, it is considered that the proposed calibrated algorithms represent the chlorophyll-a (Chl-a) and turbidity (tur) values in the Polyphytos Reservoir for the scope of this study. However, for future research, these algorithms should be validated using additional in-situ measurements, and consideration should be given to the inclusion of more diverse algorithms. Subsequently, the refined chlorophyll-a and turbidity algorithms were applied to satellite images from 2018 to 2022 to produce temporal-spatial distribution maps. For each product generated on available satellite image dates, 70 sampling points were selected throughout the reservoir, extending from the northern to the southern region. These points were plotted over a five-year time range, generating plots that illustrated the chlorophyll and turbidity profiles of the reservoir during this period. This approach facilitated a more accessible interpretation of the results through visualization, enabling comparison with other large-scale phenomena, such as fluctuations in chlorophyll and turbidity levels, variations in reservoir water level, and temperature changes. Additionally, it allowed for comparisons with sporadic measurements taken on various dates and under different environmental conditions.

3. Results and discussion

3.1. Turbidity and water level fluctuations

The left vertical axis of Fig. 2 displays the reservoir's water level fluctuations data, as measured by PCC (represented by a continuous blue line) and as calculated by the methodology proposed in this study (represented by a dashed line), spanning a period of nine years, from 2014 to the end of 2022. A comparison between the two datasets reveals that the proposed method aligns in terms of recognizing the timing of periodic fluctuations. However, concerning the absolute values, the proposed method appears to underestimate the measured water level. This discrepancy could be attributed to the accuracy of the bathymetric data at the reservoir location where the methodology has been applied. Nevertheless, this does not undermine the value of the proposed method, as it provides a relatively accurate and straightforward approach for estimating periodic fluctuations. The data consistently exhibit the repetition of a water level change pattern throughout the monitored period. This pattern is schematically illustrated in the color-coded embedded diagram in Fig. 2; occurring over an almost 12-month period, that involves water level increases (green curve) followed by decreases (red curve) and a subsequent increase (green curve) before the cycle repeats. The water level peaks towards the end of spring and reaches its lowest point during winter; this pattern is expected to be influenced by factors such as inflows (rainfall, snow melt) and discharges (irrigation, hydropower, drinking).

Moreover, the right vertical axis of Fig. 2, presented on a logarithmic scale, shows the progress of turbidity values (NTU profiles) as measured at the inlet of the TDWTP for the same nine-year period (depicted by a grey line), accompanied by a 20-day average of these recorded values (black line). The turbidity values display periodicity, with numerous maximum and minimum values observed throughout the examined period. When comparing the average turbidity profiles at the inlet of the TDWTP with water level fluctuations in Polyphytos Reservoir, it becomes evident that these values may exhibit a dependency. Specifically, during several time periods (e.g., from 12/31/2014 to 12/29/2019), a decrease in water level corresponds to a decrease in turbidity values, while an increase in water level leads to higher turbidity values.

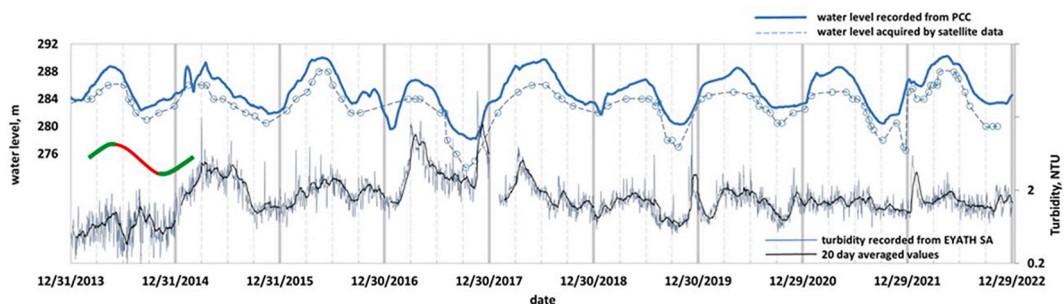


Fig. 2. The fluctuations of the turbidity values (NTU) measured in the inlet of TDWTP (grey line), the water level of Polyphytos Reservoir as identified by satellite means combined with bathymetric data (blue dashed line) and by the data provided by PCC for a period of 9 years. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The above consistency suggests a potential connection between the two phenomena (Schneider et al. 2012) which may be explained by the fact that during periods of increasing water levels, the flow velocity and turbulence in the Reservoir may increase, causing the resuspension of bottom sediments. This probably results in higher turbidity values as suspended particles increase in the water column (Li and Xia, 2023). As water levels rise, shoreline erosion intensifies, contributing more sediment to the reservoir and elevating turbidity values; however, declining water levels can reduce erosion and sediment transport, simultaneously decreasing turbidity values (Whiting, 2018).

Nevertheless, there are some periods (i.e. from 12/29/2020 till 12/29/2020) where the turbidity values are low despite significant changes in water level. This could be due to several factors such as an efficient management approach from the dam operators which could minimize the disturbance of sediments and reduce the impact of water level fluctuations on turbidity values (Kondolf et al. 2014) or due to natural processes that could take place this period of year (e.g. flocculation, sedimentation, and filtration by aquatic vegetation). The observation of a potential correlation between satellite-derived water level data and turbidity measurements is deemed significant for the optimization of water resource management and informing environmental assessments and decision-making processes.

3.2. Identification of formations in reservoir's surface

In an effort to detect any unexpected formations (anomalies) on the reservoir's surface, that could be related to deteriorated water quality at the TDWTP inlet, a comprehensive analysis of satellite images spanning a nine-year period was conducted using both the Normalized Difference NIR/Green (NDWI) (B03-B08)/(B03 + B08) and OSI indices. The application of these indices revealed multiple formations on the Reservoir over the studied period. Fig. 3 showcases two representative examples of these formations for two typical cases; the left column displays a formation captured on October 1, 2017, while the right column features a formation observed on Jan-

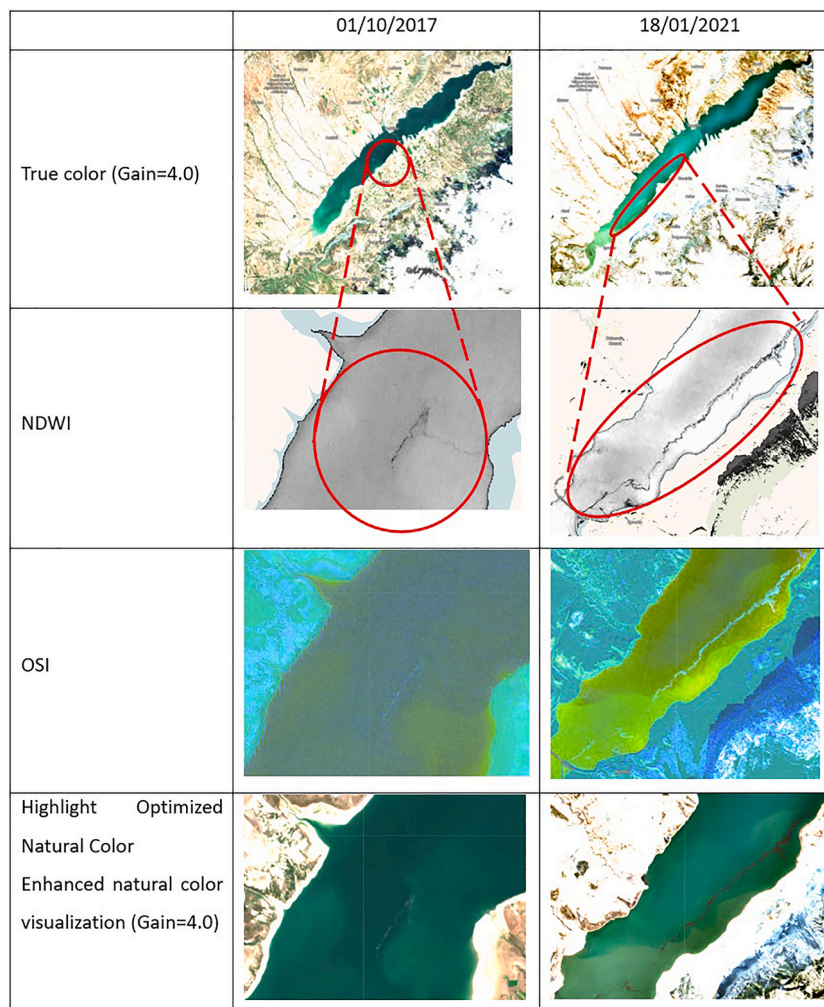


Fig. 3. Two representative reservoir formations (left: October 1, 2017; right: January 18, 2021) with location (row 1), NDWI analysis (row 2), OSI filter analysis (row 3), and visual spectrum (row 4, enhanced natural color, Gain: 2.3). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

uary 18, 2021. The first row of Fig. 3 highlights the location of the formations within the reservoir, while the second row presents the analysis results after applying the NDWI. Intriguingly, this filter effectively differentiates the formations from the water body, suggesting that the formations could indicate the presence of suspended matter or turbidity in the water, and/or a high concentration of phytoplankton biomass (Viso-Vázquez et al. 2021). The third row of Fig. 3 depicts the analysis using the OSI filter, which also distinguishes the anomalies from the main water body without adding new information.

To gain a better understanding of the origin of these formations, they were examined in the visual spectrum. The fourth row of Fig. 3 reveals how the two representative formations appear in the visual spectrum (by applying the enhanced natural color visualization, and Gain: 2.3). The observed red (right column) and green formations (left column) on the reservoir's surface could be correlated to algal blooms, suspended sediments, and the presence of hydrocarbons (as these substances have been identified in other instances in EYATH SA laboratories). A comprehensive analysis of satellite imagery, coupled with EYATH SA measurements over a nine-year period, is subsequently conducted to determine potential correlations between the identified sources and satellite based observed surface formations.

The surface was systematically scanned over a nine-year period utilizing Sentinel 2 and Landsat 8-9 repositories (300 available images with no cloud cover or satellite availability). The images were initially analyzed, to distinguish the anomalies from the water surface, using the NDWI and B02/B05 spectra for Sentinel 2 and Landsat 8-9 satellite images, respectively (anomalies have been detected for a total of 52 days). Then the formations were subsequently recorded and examined in the visual spectra, where they were categorized based on color (36 red and 16 green) and size. The formations that have been observed demonstrate a broad spectrum of sizes, ranging from several hundred to thousands of meters, and also exhibit diverse intensities, spanning from pale to intense. It is important to note that most of these formations are observed in the southern part of the Reservoir (43 of 52).

To better understand their origin and characteristics and to examine potential correlations between the presence of these formations and the water level changes, the recorded formations are presented in Fig. 4, alongside the water level fluctuations (left axis in Fig. 4) and EYATH SA measurements. The right axis of Fig. 4 presents the intensity of the identified formations on a scale of 1–3 (i.e., 1 designates the most intense and coherent formations, while 3 designates the palest formations). The formations are arranged according to their sequential emergence, colour (red or green), and dimensions, with circle points representing three unique size categories: circles with small diameters (100–200 meters), medium diameters (200–500 meters), and large diameters (over 500 meters). The green triangles in Fig. 4 indicate instances of increased hydrocarbon levels ($> 0.10\mu\text{g/l}$).

Comparison of the formation occurrence dates with the reservoir's water level fluctuation profile suggests that in most cases (45 out of 52), these formations appear when the water level begins to increase. This indicates a possible connection between the observed formations and periodic changes in the reservoir's water level. Furthermore, the first traces of hydrocarbons were recorded in January 2018, and in most cases (17 out of 21), hydrocarbon events were detected during periods of increasing water levels. Additionally, the presence of hydrocarbons often coincides with the observation of formations on the reservoir surface, with the majority being green.

Despite categorizing and classifying the formations based on their physical attributes, no additional information about their origins can be obtained. The origins and characteristics of these formations could not be discerned through on-site inspections due to late detection and insufficient records. To comprehensively investigate these phenomena and assess their potential connection to increased chlorophyll and turbidity levels, EYATH SA has commenced systematic sampling in Polyphytos Reservoir. These data will be employed to validate the proposed chlorophyll and turbidity algorithms, which are explicitly detailed in the subsequent section.

3.3. Algorithms validation

Fig. 5 (first row) shows the 30 sampling points throughout the reservoir, where water samples were collected on June 17, 2022, at 9:00 a.m., with the entire process taking approximately 2 h. The corresponding parameter values are represented using a chromatic scale. Following the analysis of the samples in the EYATH SA laboratories, the algorithm parameters were calibrated based on the Chl-a and turbidity values for the specific sampling points. The optimal fitting equation was determined by identifying and treating significantly deviating measured values as outliers, thereby ensuring a robust and accurate representation of the data. The resulting parameter values and corresponding equations are presented below:

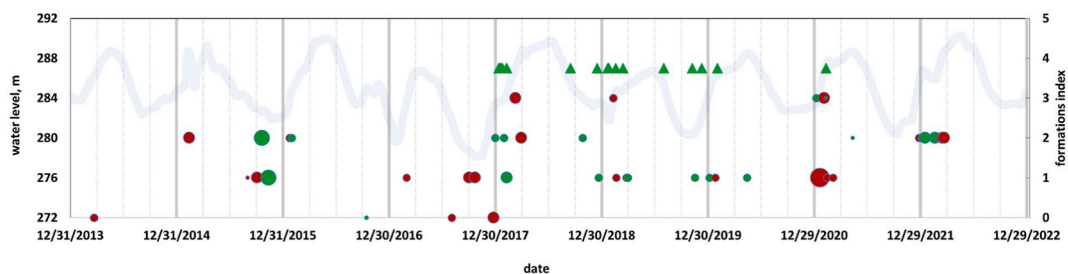


Fig. 4. The fluctuations of the water level of Polyphytos Reservoir the various formations identified on the reservoir's surface (circles stand for the corresponding red and green formations visually identified from satellite tools) and the incidents of hydrocarbons (triangles) for a 9-years period. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

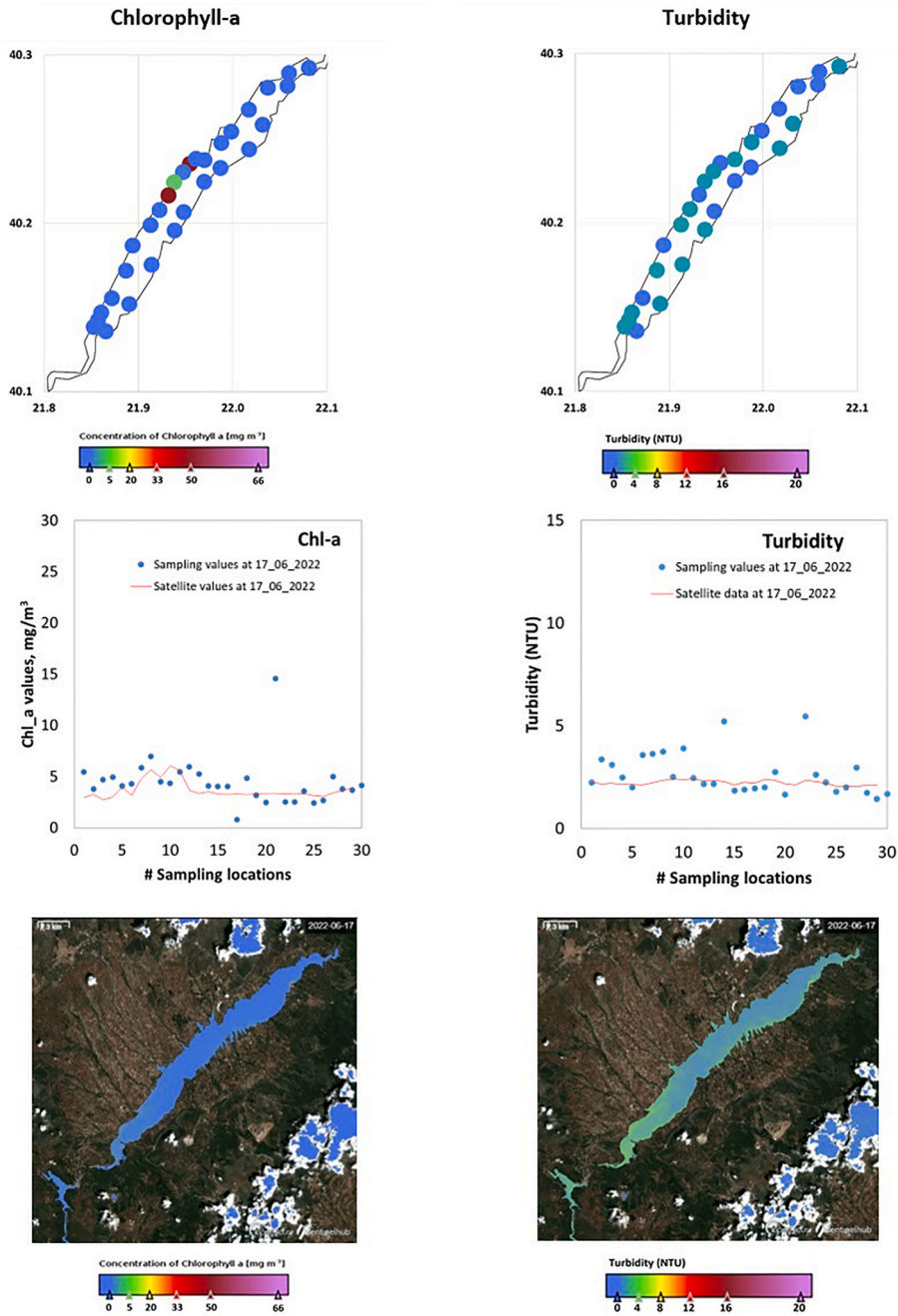


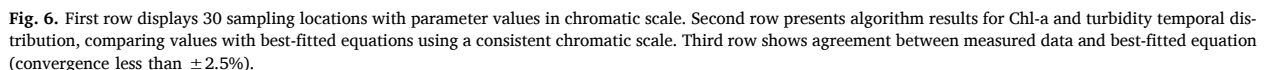
Fig. 5. The images in first row display the measured parameters (chl-a & turbidity) in each of the 30 sampling points using a chromatic scale. The images in second row show how the proposed algorithms are fitted with the measured values (convergence less than $\pm 2.5\%$). The last two images present algorithm results for Chl-a and turbidity temporal distribution.

$$Chl - a = 0.71 \left(\frac{B_{03}}{B_{01}} \right)^{3.94} \quad (3.1)$$

$$Tur = 1.98 * \frac{B_{03}}{B_{01}} - 1.42 \quad (3.2)$$

The convergence between the measured data and the best-fitted equation (presented in second row in Fig. 5) was less than $\pm 2.5\%$. Subsequently, the algorithm with the specified parameter values was employed to analyze the Satellite images to produce temporal distribution of Chl-a and turbidity values. The algorithm results are presented in third row of Fig. 5, which compares the

To evaluate the accuracy of chlorophyll and turbidity algorithms, the previously mentioned algorithms were further tested using on-site measurements from the reservoir, taken on October 4th, 2021. Unfortunately, the closest corresponding satellite imagery is dated October 5th, 2021, which represents a one-day temporal mismatch between the field measurements and satellite data. Fig. 6, row 1, displays the sampling points, along with the measured values for chlorophyll and turbidity, and the obtained values utilizing the same color scale as presented in Fig. 5. Row 2 of Fig. 6 exhibits the outcomes of the analysis derived from applying the algorithms



to the satellite images. Upon comparing the reservoir surface sampling-based measurements with the results from the satellite image analysis, a strong agreement is observed. Moreover, the third row of Fig. 6 demonstrates an excellent concordance between the reservoir surface measurements and the satellite image analysis results when juxtaposed. However, discrepancy is noted for the chlorophyll values in the southern region of the reservoir, where the satellite analysis results appear to overestimate the measured values. This deviation may be attributed to the fact that the measurements were not conducted on the same day as the satellite pass, potentially resulting in a rise in chlorophyll values on the subsequent day of measurement.

3.4. Chlorophyll and turbidity spatiotemporal footprint on Polyphytos Reservoir

In an effort to further validate the algorithms' capability to accurately estimate the reservoir's environmental conditions, they were executed for all available dates ranging from January 1, 2018, to December 31, 2022, for a total of 180 observation days. The results were then compared with the corresponding for the period available data records of EYATH SA. Figs. 7 and 8 display vertical lines representing chlorophyll concentrations and turbidity values, respectively, at 70 distinct locations throughout the reservoir. These locations span from the southernmost to the northernmost regions, corresponding to locations shown in the inset plot demonstrating the reservoir's geographical orientation while each vertical line corresponds to a day with available satellite imagery.

Regarding chlorophyll, its distribution in the reservoir is greatly variable both spatially and temporally in agreement with other studies on reservoirs around the world (e.g. Soares et al. 2012; Serra et al. 2007; Vidal et al. 2012). Based on the satellite data derived for the period 2018–2022, elevated chlorophyll values ($>60 \mu\text{g L}^{-1}$) are usually observed in the shallow southern regions of Polyphytos Reservoir whereas the deeper northern regions of the reservoir exhibit generally lower chlorophyll values, as indicated by the blue hue on the chromatic scale (Fig. 7). This spatial variability was evident also in the determined values of chlorophyll-a infield samples collected and analyzed by EYATH SA during 2021–2022 (Figs. 5 and 6). During this period, especially high Chl-a values were recorded in the southern part of the reservoir, exceeding $150 \mu\text{g L}^{-1}$ on January 4th and 20th, while Chl-a concentrations were substantially lower ($5\text{--}30 \mu\text{g L}^{-1}$) in the central and northern parts of the it. Microscopic analysis and phytoplankton measurements at the same time, revealed during this time the development of a dense cyanobacterial bloom in the reservoir (Katsiapi et al. in prep). Cyanobacterial blooms are well-known for demonstrating high spatial variability while developing in a water body (Sharp et al. 2021). This phytoplankton patchiness along the reservoir may be attributed to differences in the nutrient distribution and availability, light availability and stratification (Serra et al., 2007) while the water retention time and river inflow may as well in turn strongly influence phytoplankton communities and their growth (Soares et al., 2012).

Regarding the temporal distribution of Chl-a, variability is also observed both within the same year and between the years (Fig. 7). For example, within the same year, higher values (average values of the 70 locations) are usually observed during colder months (November–March) (the water level of the reservoir is typically kept lower than in summer time due to management), such as the winters of 2018, 2021, and 2022. In 2019 and 2020, the highest chlorophyll values (average values of the 70 locations) were observed during summertime. This temporal variability is largely the result of the dominance of different algal groups in the reservoir's phytoplankton; this dominance in turn depends on several factors such as light, nutrients, water retention time, species composition etc (Desortova 1981). In the long term, this temporal variability will also be subjected to climate change which is associated with higher water temperature and stable thermal conditions, thus is expected to promote algal growth, particularly cyanobacterial blooms (Kraemer et al., 2015).

As it has previously been suggested (Fig. 2, paragraph 3.1), the turbidity levels in the inlet of TWDTP are significantly affected by the water level fluctuations. It is well known that the management and dam operational procedures can influence water movement and sediment distribution by causing localized turbulence and resuspension of sediments within the reservoir (Schneider et al. 2012). Fig. 8a shows that the southern areas of Polyphytos Reservoir systematically have higher turbidity values, while the northern areas are rarely characterized by turbidity values above 8 NTU. Since Fig. 8a suggests that the southern areas have higher turbidity values, the observed behavior of the overall increase of turbidity in the TDWTP inlet could be attributed to the increase of turbidity at this specific area. Moreover, the second bottom part of Fig. 8b illustrates the average turbidity values derived from satellite analysis (logarithmic scale on the left vertical axis), and daily turbidity values recorded at the TDWTP input (logarithmic scale on the right vertical axis). The ratio between the values of the two axes is $\sim 1/10$. The anticipated discrepancy between satellite-derived turbidity in the

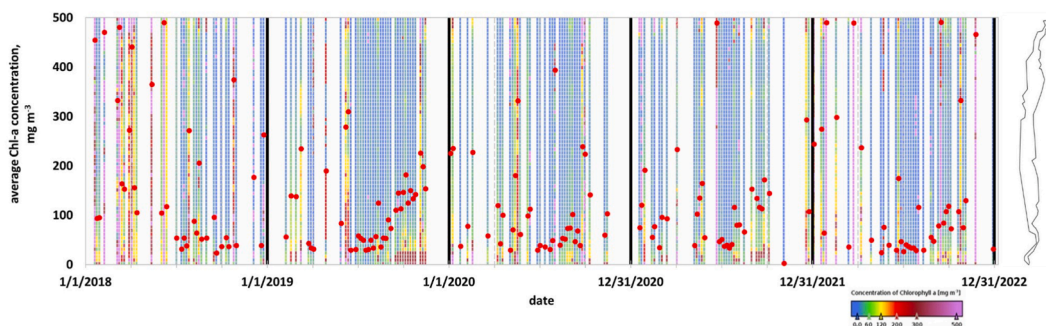


Fig. 7. Vertical representation of chlorophyll concentrations at 70 distinct locations throughout the reservoir from January 1, 2018, to December 31, 2022. Each vertical line signifies a day with available satellite imagery. Locations span from the southernmost to the northernmost regions, as depicted in the inset plot showing the reservoir's geographical orientation.

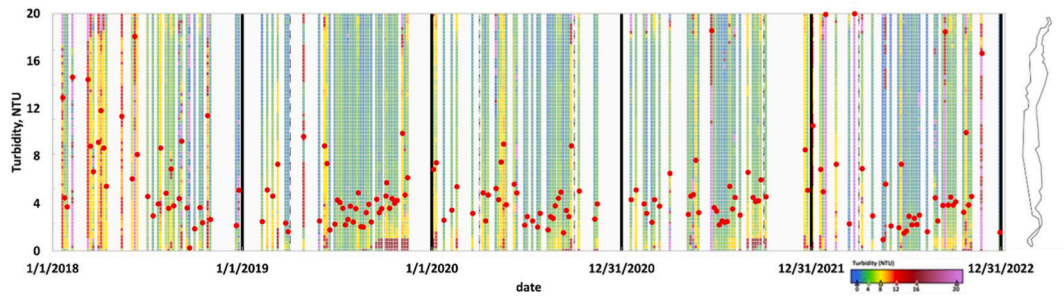


Fig. 8a. Vertical representation of Turbidity values at 70 distinct locations throughout the reservoir from January 1, 2018, to December 31, 2022. Each vertical line signifies a day with available satellite imagery. Locations span from the southernmost to the northernmost regions, as depicted in the inset plot showing the reservoir's geographical orientation.

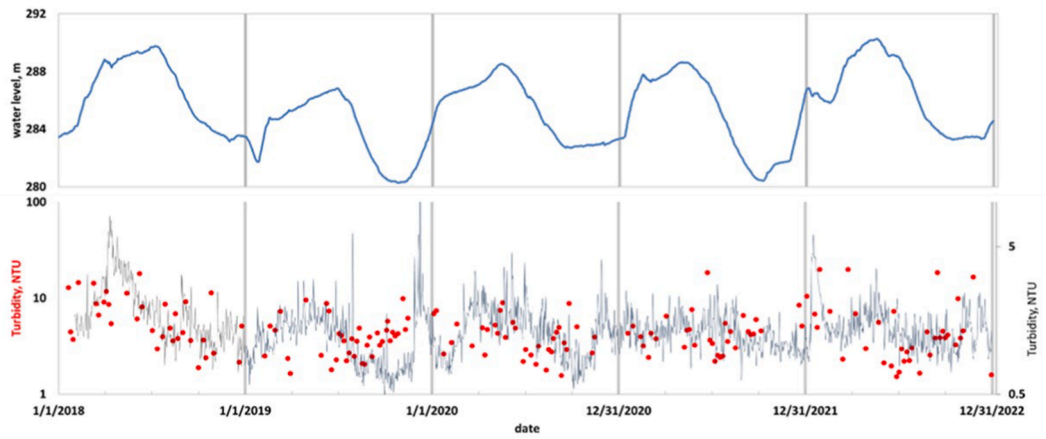


Fig. 8b. Comparison of average turbidity values from satellite analysis (logarithmic scale on the left vertical axis) with daily turbidity values recorded at the TDWTP input (logarithmic scale on the right vertical axis).

reservoir and turbidity values at the inlet of TDWTP can be attributed to numerous physical, chemical, and biological processes, such as settling and sedimentation, flocculation and coagulation, filtration, biological activity, and hydrodynamic processes, which are expected to decrease turbidity within downstream reservoirs and channels and contribute to particle removal (Davies-Colley and Smith, 2001).

3.5. Workflow

The present analysis has shown that the use of specific algorithms, such as Chl-a, Tur and OSI, in evaluating satellite imagery can effectively assess the water quality of a drinking water reservoir. The algorithms used in this study offer a reliable representation of the probability that the water quality at designated positions in the reservoir is either satisfactory or inadequate. Moreover, the methodological steps used in this study can be summarized in proposing a systematic approach (presented in Fig. 9) for identifying unforeseen reservoir surface formations and evaluating water quality, which could be incorporated into water quality control procedures by organizations concerned with rapid environmental assessments of large water bodies. Specifically, preliminary inspection using the NDWI algorithm can provide initial insights into the surface water formations and prompt preventative actions or urgent sampling campaigns. Advanced analysis with Se2WaQ and OSI algorithms (applied on Sentinel 2A images which incorporate atmospheric correction services) can provide extensive information about the environmental status of the reservoirs. The proposed workflow in Fig. 9, can aid in detecting unusual formations, assessing surface water quality, optimizing sampling procedures, and improving the identification of pollutants in free surface water. The integration of this workflow into water utility procedures could enhance water treatment processes.

For a water utility like EYATH SA, incorporating satellite-based monitoring tools into operational procedures can be hindered by technical, financial, and organizational challenges. Moreover, the processing and interpreting satellite data may require specialized expertise that is not readily available within water utilities (Katsanevakis et al., 2012). Additionally, the integration of satellite-derived information with existing monitoring systems may necessitate the development of new data processing workflows and methods. However, the present study proposes and validates a methodology that could be applied by any water utility or an organization that has a minimum expertise by using information that is available in public accessed browsers and scripts. While acknowledging the inherent complexities of water bodies that pose significant challenges in interpreting remote sensing outcomes, this study also underscores the crucial need for water utilities, as end-users, to accurately discern the origin of observed formations. The spectral signatures captured from satellite imagery can be influenced by several water constituents including, but not limited to, phytoplankton,

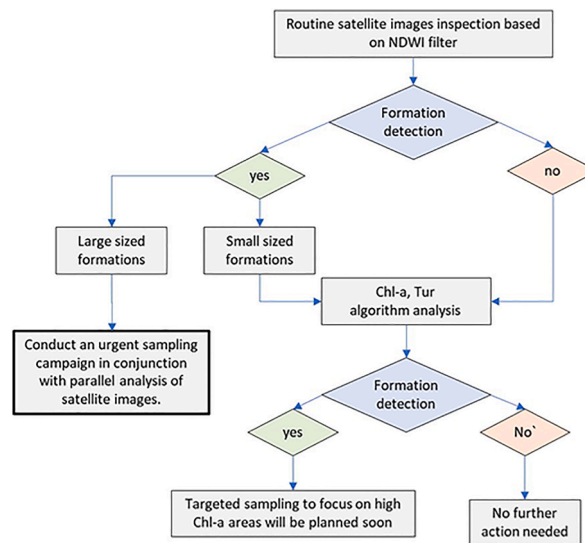


Fig. 9. The workflow proposed to incorporate satellite remote sensing into the operational procedures of an end user having medium to low expertise at satellite remote sensing tools.

suspended solids, and dissolved organic matter (Giardino et al., 2007). Such intricacies necessitate detailed in-situ measurements and interpretation to accurately extract valuable information from the data. Given this complexity, it is vital to foster an interdisciplinary collaboration between water utilities and domain experts to enhance the collective understanding of remote sensing applications. Such partnerships can facilitate the development of more sophisticated, accurate, and user-friendly remote sensing tools specifically tailored to the needs of water quality monitoring. This collaborative approach not only leverages the technical expertise of specialists but also integrates the practical insights of end-users, thus driving the creation of more effective and implementable solutions for water quality assessment and management.

4. Conclusion

The proposed methodology, which employs freely available tools, is motivated by the need to assess the water quality of Polyphytos Reservoir and is accessible to end-users who may not have extensive knowledge in satellite monitoring. The analysis of over 300 satellite images, coupled with the use of algorithms such as NDWI, Se2WaQ, and OSI, has proven effective in detecting potential formations that could impact surface water quality and characterizing water quality over a nine-year period. Additionally, the methodology proposed to determine fluctuations in the water level of the reservoir could be useful when relevant data is not available. The overall approach has enabled the identification of vulnerable areas and timeframes within the reservoir, thereby aiding in the optimization of the sampling procedure and the prevention of potential issues in water purification. Specifically, the Chl-a and Tur algorithms suggest that two regimes exist in the reservoir (south and north) with distinct qualitative characteristics that change throughout the year and are influenced by various parameters, such as dam management concerning water level fluctuations and environmental factors like temperature.

Further understanding these interactions is essential for effective water resource management and addressing environmental concerns. The proposed methodology is designed to generate chlorophyll and turbidity footprints over extended time periods and can be effectively applied by end users with minimal expertise. This approach enables the automatic creation of analogous footprints that represent temporal concentration variations for any parameter within any inland water system, spanning any desired time frame. Future research endeavors should concentrate on evaluating the potential of various types of algorithms (i.e. red/green bands) in conjunction with an expanded range of in-situ measurements. This would enhance the validation of these algorithms and refine their predictive capabilities and contribute to the development of an integrated water quality monitoring system using satellite instruments, which could be successfully incorporated into the operational processes of water utilities.

Ethical statement for solid state ionics

Hereby, I/insert author name/consciously assure that for the manuscript/insert title/the following is fulfilled:

- 1) This material is the authors' own original work, which has not been previously published elsewhere.
- 2) The paper is not currently being considered for publication elsewhere.
- 3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
- 4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
- 5) The results are appropriately placed in the context of prior and existing research.

- 6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference.
- 7) All authors have been personally and actively involved in substantial work leading to the paper, and will take public responsibility for its content.

The violation of the Ethical Statement rules may result in severe consequences.

To verify originality, your article may be checked by the originality detection software iThenticate. See also <http://www.elsevier.com/editors/plagdetect>.

I agree with the above statements and declare that this submission follows the policies of Solid State Ionics as outlined in the Guide for Authors and in the Ethical Statement.

Author contributions statement

Dr. John Lioumbas: Conceptualization, Methodology, Software, Validation, Formal Analysis, Writing - Original Draft, Project Administration, Supervision.

Aikaterini Christodoulou: Writing - Review & Editing, Methodology, Validation, Data Curation.

Dr. Matina Katsiapi: Writing - Review & Editing, Methodology, Validation, Formal Analysis, Investigation and Resources.

Dr. Nikoletta Xanthopoulou: Writing - Review & Editing, Methodology, Validation, Formal Analysis and Resources.

Dr. Panagiota Stournara: Writing - Review & Editing, Data Curation and Visualization.

Thomas Spahos: Investigation and Resources.

Dr. Georgia Seretoudi: Supervision and Resources.

Dr. Alexandros Mentis: Supervision and Resources.

Nopi Theodoridou: Supervision and Project Administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rsase.2023.101042>.

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